Pre-Analysis Plan:

Variables: Which variables will you include? How will these variables be coded— will you use them the way they are or recode them, and if so, how?

- I would include a dummy variable indicating whether the respondent has at least one daughter (coded as 1) or not (coded as 0).
- As control variables, I would include a dummy for female, age (continuous), education level, marital status, number of children, possibly regional indicators, and year fixed effects.
- All variables should be recoded carefully, as missing values are often coded as negative numbers in the dataset.
- Most importantly, we need to identify suitable dependent variables that accurately capture the concept of gender equality, as this is crucial for the analysis.

• Inclusion/exclusion criteria: Are there any observations you will exclude? If yes, which? Which years will you include/what period will you study?

- I would like to include all available years for which data exist. However, some years may contain missing information due to incomplete responses or challenges in constructing key variables.

• Exact analysis: Please write down the exact specification of the regression model, including the names of the dependent and independent variables, and whether you use robust standard errors or something else.

In case of probit:

$d_i = 1(X'\beta + \phi_1 maritalstatus_i + \gamma_i education_i + \Pi_i location_i + \rho experience_i +$
$\ln(income)i + \delta daughter_i + \varepsilon_i > 0)$

X: individual characteristics: gender, age etc.

- I would prefer to create a time-dependent variable for having a daughter in order to identify when individuals have a daughter and how their attitudes or behaviors change thereafter. However, the given hypotheses require only a binary indicator for whether individuals have at least one daughter, which simplifies the analysis but may limit the ability to detect more nuanced effects.

- The choice of regression model will depend on the nature of the dependent variable. For ordinal outcomes—such as attitudes toward gender equality measured on a 1–4 or 1–5 scale where I would use an ordered probit model. If the dependent variable has a wider range (e.g., 0–10), an OLS regression would be appropriate. For binary outcomes, I would consider using probit or logit models, depending on the distribution and theoretical justification.

- In all cases, I would use robust standard errors to account for heteroskedasticity and improve the reliability of the estimated coefficients.

- Hypothesis 1: To measure support for gender equality in attitudes, I will use the variable plh0308_v2, which reflects agreement with the statement "It is best if men and women work the same amount so that they can share responsibilities", measured on a 7-point Likert scale. I will estimate an ordered probit regression with robust standard errors, including standard control variables as described above (e.g., gender, age, education, marital status, number of children, year and region fixed effects). I expect the coefficient for having at least one daughter to be positive, indicating greater support for gender equality.

- Hypothesis 2: Political orientation will be captured using the variable plh0004, which ranges from 0 (far left) to 10 (far right). Since higher values indicate more right-leaning views, I expect the coefficient on having a daughter to be negative, suggesting a shift toward more left-leaning political preferences. I will use OLS regression (oprobit could be also used) with robust standard errors for this analysis.

- Hypothesis 3: To measure gender equality in household behavior or division of labor, I will use the variable plh0308_v1 ("Men and women should be equally involved in both work and household/family responsibilities", FID 2012). While I acknowledge that this variable is not an ideal behavioral measure, it is the most relevant item available. I will generate a binary variable from this item and estimate a probit regression with robust standard errors.

- Hypothesis 4: This hypothesis requires an examination of interaction effects between gender and daughter presence. Specifically, I will include an interaction term (female ×

daughter) in the model to compare the differential effects for men and women. To assess whether the effect of having daughters differs significantly between male and female respondents, I will perform a joint significance test on female and female × daughter (e.g., via a Wald test).

• Interpretation of results: Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test result is in line with the hypothesis or not?

- To evaluate whether the results support the hypotheses, I would use hypothesis testing techniques, such as Wald tests or t-tests, depending on the model specification and the nature of the coefficients being tested.